Nümerik integraller

Belirli integraller

$$\int_a^b f(x)\,dx = \sum_{k=1}^{n} \alpha_k f(x_k) \quad \text{"quadrature"}$$

$x_0, x_1, x_2, \dots, x_n \in [a,b]$

Lagrange IP $\quad P_n(x) = \sum_{k=0}^{n} L_{n,k}(x) f(x_k)$

$$L_{n,k}(x) = \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{(x-x_i)}{(x_k-x_i)}$$

$f \in C^{n+1}[a,b]$

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{i=0}^{n} (x-x_i) \qquad \xi = \xi(x) \in (a,b)$$

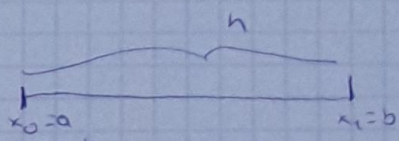
$$\int_a^b f(x)\,dx = \int_a^b \sum_{k=0}^{n} L_{n,k}(x) f(x_k)\,dx + \int_a^b \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x-x_i)\,dx$$

(1. terim: Tahmin; 2. terim: Hata)

$$= \underbrace{\sum_{k=0}^{n} \alpha_k f(x_k)}_{\text{Tahmin}} + \underbrace{E(f)}_{\text{Hata}}$$

$$\alpha_k = \int_a^b L_{n,k}(x)\,dx$$

Trapezoid (yamuk) kuralı



$$f(x) \simeq P_1(x) = \frac{x-x_1}{x_0-x_1} f(x_0) + \frac{x-x_0}{x_1-x_0} f(x_1)$$

$$E_1(x) = \frac{f''(\xi(x))}{2} (x-x_0)(x-x_1) \qquad \xi \in (x_0, x_1)$$

$$\int_a^b f(x)\,dx = \int_a^b P_1(x)\,dx + \int_a^b E_1(x)\,dx$$

İntegral için ortalama (ara) değer teorimi

$f \in C[a,b]$ ve $g(x)$ $a \le x \le b$ integrallenebilir ve işaretini korur öyle bir $c \in (a,b)$ vardır ki, $\int_a^b f(x) g(x)\,dx = f(c) \int_a^b g(x)\,dx$

$$\int_{x_0}^{x_1} \underbrace{f''(\xi(x)) (x-x_0)(x-x_1)}_{g(x)\ (\text{işaret değiştirmiyor})}\,dx$$

$$= f''(\xi_1) \int_{x_0}^{x_1} (x-x_0)(x-x_1)\,dx \;\Big|_{h = x_1 - x_0} = \frac{-h^3}{\quad} f''(\xi_1)$$

$$= \frac{h^3}{6} f^{(n+1)}$$

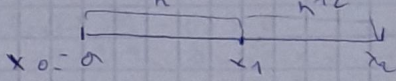
$$\int_a^b f(x)\,dx = \int_a^b P_1(x)\,dx + \int_a^b E_1(x)\,dx$$

$$= \left[\frac{(x-x_1)^2}{2(x_0-x_1)} f(x_0) + \frac{(x-x_0)^2}{2(x_1-x_0)} f(x_1) \right]_{x=x_0}^{x=x_1}$$

$$+ -\frac{h^3}{12} f''(\xi_1)$$

$$= \frac{h}{2}\left[f(x_0) + f(x_1) \right] - \frac{h^3}{12} f''(\xi_1)$$

Simpon kurali



$$P_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1)$$

$$+ \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

$$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} P_2(x)\,dx + \int_{x_0}^{x_1} E_2(x)\,dx$$

ordinan diferansiyel Taylor

$$f(x) = f(x_1) + f'(x_1)(x-x_1) + \frac{f''(x_1)}{2!}(x-x_1)^2 + \frac{f'''(x_1)}{3!}(x-x_1)^3$$

$$+ \frac{f^{(4)}(\xi)}{4!}(x-x_1)^4$$

$$\int_a^b f(x)\,dx = \left[f(x_1)(x-x_1) + \frac{f'(x_1)}{2}(x-x_1)^2 + \frac{f''(x_1)}{6}(x-x_1)^3 \right.$$

$$\left. + \frac{f'''(x_1)}{24}(x-x_1)^4 \right]_{x_0}^{x_2} + \int_{x_0}^{x_2} \frac{f^{(4)}(\xi(x))}{24}(x-x_1)^4\,dx$$

$$\frac{f^{(4)}(\xi_1)}{120}\, 2h^5$$

$$f''(x_1) = \frac{1}{h^2}\left[f(x_0) - 2f(x_1) + f(x_2) \right] - \frac{h^2}{12} f^{(4)}(\xi_2)$$

$$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} P_2(x)\,dx + \int_{x_0}^{x_2} E_2(x)\,dx = 2h f(x_1) + \cdots$$

$$\frac{h^3}{3}\left[\frac{1}{h^2}\left[f(x_0) - 2f(x_1) + f(x_2) \right] - \frac{h^2}{12} f^{(4)}(\xi_2) \right]$$

$$+ \frac{f^{(4)}(\xi_1)\, h^5}{60}$$

$$= \frac{h}{3}\left[f(x_0)+4f(x_1)+f(x_2)\right] - \frac{h^5}{12}\left[\frac{1}{3}f^{(4)}(\xi_2) - \frac{1}{5}f^{(4)}(\xi_1)\right]$$

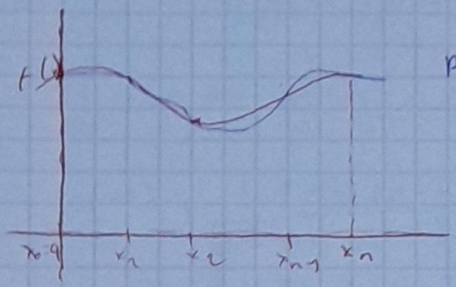
$$\int_a^b f(x)\,dx = \underbrace{\frac{h}{3}\left[f(x_0)+4f(x_1)+f(x_2)\right]}_{\text{Tanımı}} - \underbrace{\frac{h^5}{90}f^{(4)}(\xi)}_{\text{Hata}}$$

Kapalı Newton-Cotes formülü (Genelleme)

$[a,b]$ kapalı aralık

$x_0 = a$, $x_n = b$

$x_k = x_0 + kh$ $\qquad h = \frac{b-a}{n}$ $\qquad k = 0, 1, \dots, n$



$$P_n(x) = \sum_{k=1}^{n} f(x_k) \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{(x-x_i)}{(x_k-x_i)}$$

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!}\prod_{k=0}^{n}(x-x_k)$$

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx = \sum_{k=0}^{\infty} \alpha_k f(x_k)$$

$$\alpha_k = \int_a^b \prod_{\substack{i=0 \\ i\neq k}}^{n} \frac{x-x_i}{x_k-x_i}\,dx$$

Teorem $\sum_{k=0}^{n} \alpha_k f(x_k)$ $(n+1)$ adet noktadan kapalı NC formülü olsun

$x_0 = a$, $x_n = b$ ve $h = \frac{b-a}{n}$

$n=1 \to$ Trapezoid

$n=2 \to$ Simpson

Böylesi $\xi \in (a,b)$ sayısı, n çift ve $f \in C^{n+2}[a,b]$ olacak biçimde

$$\int_a^b f(x)\,dx = \sum_{k=0}^{n} \alpha_k f(x_k) + \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!}\int_0^n t^2(t-1)\cdots(t-n)\,dt$$

n tek ve $f \in C^{n+1}[a,b]$

$$+ \frac{h^{n+1} f^{(n+1)}(\xi)}{(n+1)!}\int_0^n t(t-1)\cdots(t-n)\,dt$$

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formülleri de geçerlidir.

Açık Newton - Cotes formülü (Genelleme)

(a,b) açık aralık

$x_{-1} = a$, $x_{n+1} = b$

$x_k = x_0 + kh \qquad h = \frac{b-a}{n+2}$

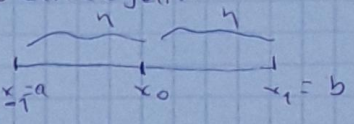
$k = 0, 1, \dots, n$

$x_0, x_1, \dots x_n \in (a,b)$



$n = 0$ durumunda formülü belirleyelim

$n = \frac{b-a}{2} \qquad x_0 = a + h$



$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2}(x - x_0)^2$$

$$\int_a^b f(x)\,dx = \int_{x_{-1}}^{x_1} f(x_0)\,dx + \int_{x_{-1}}^{x_1} f'(x_0)(x - x_0)\,dx + \frac{1}{2}\int_{x_{-1}}^{x_1} f''(\xi(x))(x - x_0)^2 dx$$

$$= f(x_0)(x_1 - x_{-1}) + f'(x_0)\frac{(x - x_0)^2}{2}\Big|_{x_{-1}}^{x_1} + \frac{1}{2} f''(\xi_1)\int_{x_{-1}}^{x_1} (x - x_0)^2 dx$$

$$= 2h f(x_0) + \frac{h^3}{3} f''(\xi_1)$$

Uygulama

1) $\int_{0,5}^{1} x^4 \, dx = \frac{31}{160} = \frac{31}{160} = 0,1935$ 2) $\int_{e}^{e+1} \frac{1}{x^{\ln x}} \, dx \approx 0,2225$

integrallerin değerini Trapezoidal ve Simpson formülü kullanarak tahmini hesaplayınız

1) $f(x) = x^4$, $\underbrace{0,5}_{a} \le x \le \underbrace{1}_{b}$

$x_0 = 0,5$
$x_1 = 1$
$h = b - a = 1 - 0,5 = 0,5$

$$\int_{0,5}^{1} f(x) \, dx \approx \frac{h}{2} [f(x_0) + f(x_1)]$$

$$\int_{0,5}^{1} x^4 \, dx = \frac{0,5}{2} [(0,5)^4 + 1^4] = \frac{1}{4} \left[\frac{1}{16} + 1\right] = \frac{17}{64} \approx 0,265$$

simpson

$$\int_{a}^{b} f(x) \, dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$x_0 = 0,5$ $x_1 = x_0 + h$
$x_2 = 1$ $n = \frac{b-a}{2}$

$h = \frac{1 - 0,5}{2} = \frac{1}{4} = 0,25$

$1 = 0,5 + 0,25 + 0,25$

$$\frac{1/4}{3} \left[\left(\frac{1}{2}\right)^4 + 4\left(\frac{3}{4}\right)^4 + 1^4\right] = \frac{1}{12} \left[\frac{1}{16} + \frac{81}{64} + 1\right] = \frac{149}{1264} = 0,1340$$

$x_0 = e$ $x_1 = x_0 + h$
$x_2 = e + 1$ $h = \frac{b-a}{2}$

$h = \frac{e + 1 - e}{2} = \frac{1}{2}$

$x_1 = e + \frac{1}{2} = e + 0,5$

$$\frac{1}{6} \left[\frac{1}{e^{\ln e}} + 4 \frac{1}{\left(\frac{1}{2} + e\right) \ln\left(\frac{1}{2} + e\right)} + \frac{1}{(1+e)\ln(1+e)}\right] \approx 0,227$$

$x_0 = e$, $x_1 = 1+e$ $\quad h = b-a = 1$

$$\int_e^{e+1} \frac{1}{x \ln x}\, dx \approx \frac{1}{2}\left[\frac{1}{e \ln e} + \frac{1}{(1+e)\ln(1+e)}\right] \approx 0,2503$$

3) Ara nokta formülünü kullanarak

$$\int_0^{\pi/4} \sin x\, dx = 1 - \frac{1}{\sqrt{2}} \approx 0,2929$$

bu integrali tahmini hesaplayınız

$$\int_a^b f(x)\,dx \approx 2h\, f(x_0), \quad h = \frac{b-a}{2} \qquad a = 0$$

$$x_0 = a + h \qquad b = \frac{\pi/4 - 0}{2} = \pi/8$$

$$x_0 = 0 + \pi/8 = \pi/8$$

$$\int_0^{\pi/4} \sin x^2\, dx = 2\frac{\pi}{8} \cdot \sin(\pi/8)$$

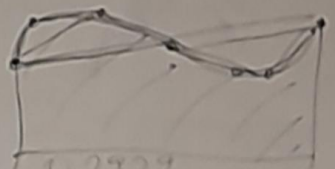
$$= \frac{3,14}{4} \cdot \sin\left(\frac{180}{8}\right) \approx 0,3005$$

4) $\int_{0.5}^{0} x \ln(1+x)\,dx$ Trapezoidal, Simpson ve Romberg formülleri kullanarak hesaplayın

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_1)\right] - \frac{h^3}{12} f''(\xi)$$ Trap.

3) Ara nokta formülü kullanarak

$$\int_0^{\pi/4} \sin x \, dx = 1 - \frac{1}{\sqrt{2}} \approx 0{,}2929$$



$$\int_0^{\pi/4} \sin x \, dx = 2\,\frac{\pi}{8}\,\sin(\pi/4)$$

$\pi/4$

bu integrali tahmini hesaplayın

$$\int_a^b f(x)\,dx \cong 2h\, f(x_0)$$

Ara nokta, $h = \frac{b-a}{2}$

$x_0 = a + h$

$a = 0$

$b = \pi/4$

$h = \frac{\pi/4 - 0}{2} = \pi/8$

$x_0 = 0 + \pi/8 = \pi/8$

$$= \frac{3\pi}{4}\,\sin\left(\frac{180^\circ}{8}\right) \text{ m}$$

$\approx 0{,}3005$

4) $\int_{0.15}^{0} x \ln(1+x)\, dx$

Trapezoidal, Simpson ve Romberg formülleri kullanarak hesaplayın.

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[f(x_0)+f(x_1)\right] - \frac{h^3}{12} f''(\xi)$$ Trapezoidal ✓

(tahmini değer; hata)

$$= \frac{h}{3}\left[f(x_0) + 4f(x_m) + f(x_2)\right] - \frac{h^5}{90} f^{(4)}(\xi)$$ Simpson. ✓

(hata)